

KRATI JOSHI

Backend Engineer | Node.js • TypeScript • PostgreSQL • Kafka

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PROFESSIONAL SUMMARY

Backend Engineer with 2+ years building scalable, production-grade systems for enterprise and government clients. At SAI Computers Limited, shipped 120+ REST APIs powering SMRITI 3.0, an AI-analytics utility platform deployed across 4 government DISCOMs. Skilled in RBAC design, multi-tenant architecture, microservices, and event-driven pipelines.

EXPERIENCE

Backend Developer — SAI Computers Limited Feb 2024 – Present

- **120+ production REST APIs:** Designed and maintained 120+ production REST APIs using Node.js, TypeScript & Express.js serving 4 government DISCOMs.
- **RBAC & Auth:** Built JWT authentication + dynamic RBAC framework covering 8+ roles with granular feature-level permissions, achieving zero unauthorised-access incidents.
- **Performance optimisation:** Cut average API response time by ~40% via in-memory caching strategies, PostgreSQL index tuning, and query rewrites on high-volume reporting endpoints.
- **AI Analytics integration:** Integrated SaiGPT (NL-to-SQL) enabling non-technical business users to query operational data in natural language, reducing analyst turnaround from days to minutes.
- **Event-driven pipelines:** Architected Kafka producer-consumer pipeline for asynchronous processing, handling 10,000+ field images/day with dead-letter queue failure handling.
- **Multi-tenant backend:** Engineered a shared-schema multi-tenant system with middleware-enforced org-scoping, ensuring zero cross-tenant data leakage.

KEY PROJECTS

SMRITI 3.0 — Enterprise Utility Management & AI Analytics Platform

Tech: Node.js, TypeScript, Express.js, PostgreSQL, SaiGPT, AWS S3

- Designed multi-tenant org management system adopted by PVVNL, DVVNL, Tripura & Meghalaya DISCOMs, supporting 1,000+ concurrent users across shared infrastructure.
- Built configurable Widget Framework: analysts configure custom SQL-backed KPI widgets without any developer intervention.
- Achieved ~35% faster dashboard load times via a cache-aside pattern with intelligent key-level invalidation on data mutations.
- Designed RBAC with feature-level granular permissions; zero cross-tenant data leakage verified by automated integration tests.

APMC Mandi Portal — Gujarat Government Trading & Licensing Platform

Tech: Node.js, Express.js, CouchDB, RBAC, JWT, SBI Payment Gateway, eSign

- Led backend development for a multi-role government trading and licensing platform serving traders, panel members, and licensing officers.
- Designed middleware-enforced RBAC and role-specific workflows covering submission, correction, approval, and re-validation stages.
- Built role-specific dashboards and approval flows while preventing data duplication or loss during corrections and resubmissions.
- Integrated SBI payment gateway and eSign for licensing fee collection and digital document authentication, replacing a paper-based process.

TECHNICAL SKILLS

Languages & Frameworks: Node.js, TypeScript, JavaScript, SQL, Express.js, Prisma, Joi, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, CouchDB

Architecture:

REST APIs, Microservices, RBAC, JWT Auth, Multi-Tenant Systems, Kafka, API Security, System Design

Tools & Integrations:

Git, AWS S3, Postman, SBI Payment Gateway, eSign

EDUCATION

B.Tech Computer Science (Artificial Intelligence) — AKTU (GPA: 8.01/10 80.12%)	2020 – 2024
12th (Senior Secondary) — UP Board (77%)	2020
10th (Secondary) — UP Board (87.5%)	2018